

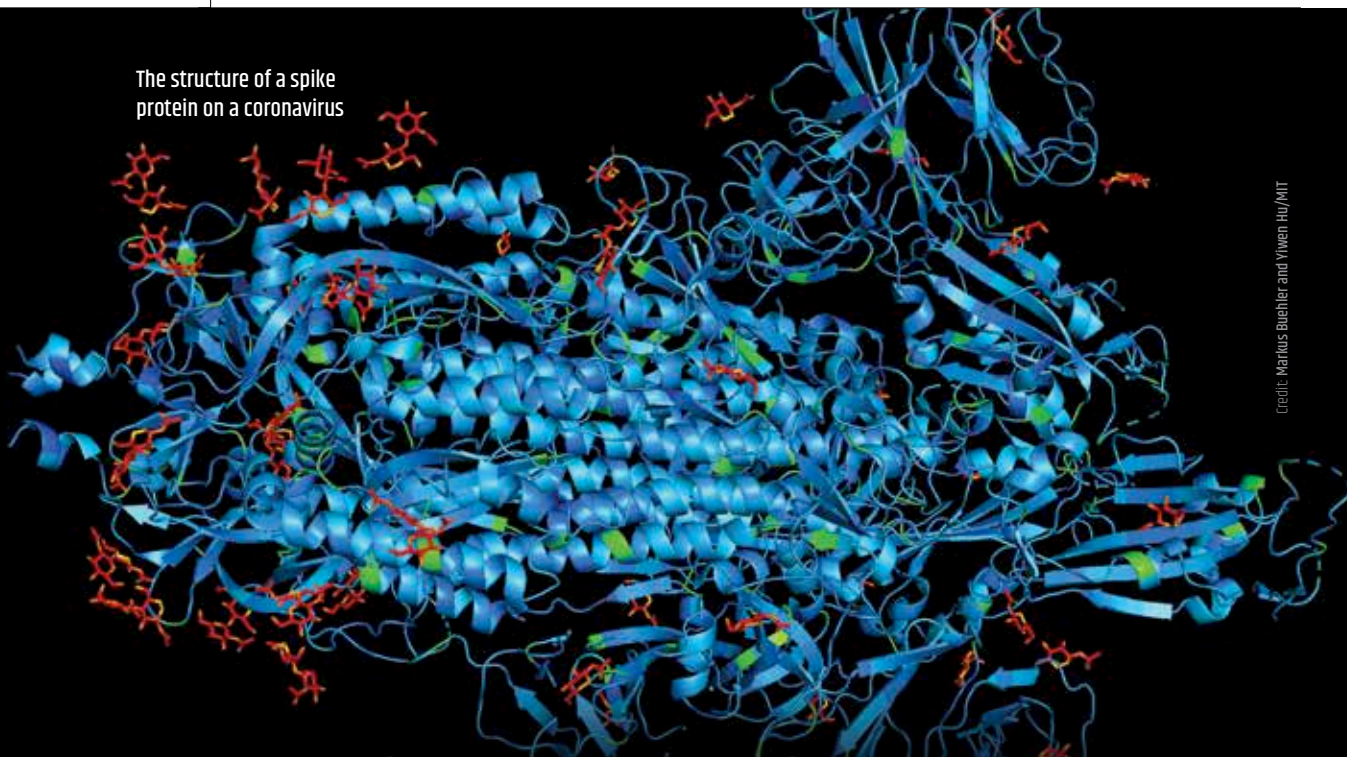
**COVID-19 and our Microbiomes**

Here's how you could develop a healthy immune system while being locked-up at home. <https://digit.in/dec20-19>

**New Breakthrough**

Oxford University and AstraZeneca developed a COVID-19 vaccine showing upto 90% efficacy. <https://digit.in/dec20-20>

The structure of a spike protein on a coronavirus



Credit: Markus Buehler and Yiyen Hu/MIT

FIGHTING THE PANDEMIC

PART IV

The research directions to stop the spread of the novel coronavirus

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ver the past month, several new vaccines began development, and as compared to 213 last month, there

are now 240 vaccines in various stages. Three more vaccines have entered clinical trials. There is consid-

erable research being done on ways to increase the efficiency of any vaccine that enters into production, including orally administered immunostimulants, adding nanoparticles that mimic antigens, and drugs that prime the immune response. We are still figuring out certain fundamental aspects about the virus, such as whether it is more infectious in certain weather conditions or how exactly it can gain entry into the cell. As self diagnosis and appropriate behaviour both play a role in containing the spread, low cost kits that allow people to test at home within short periods of time before venturing out are another area of active research.

HOW THE VIRUS SPREADS

One of the persistent rumours about the coronavirus in the early period of the pandemic was that just like

the viruses that cause the common cold, SARS-CoV-2 is less likely to spread when the temperatures were warmer and more humid. Princeton researchers who had experience in studying the dynamics of infectious diseases and the weather, ran simulations back in June 2020 to check how different weather conditions would affect the spread of the virus. Since they did not know at that time if the virus spread better in warm temperatures or not, they ran three different scenarios based on the properties of known coronaviruses, including the variants that cause the common cold. The study showed that the behaviour of the people was the most important factor in the spread of the virus, and the weather hardly played a role because most of the population was not immune to the virus. The weather